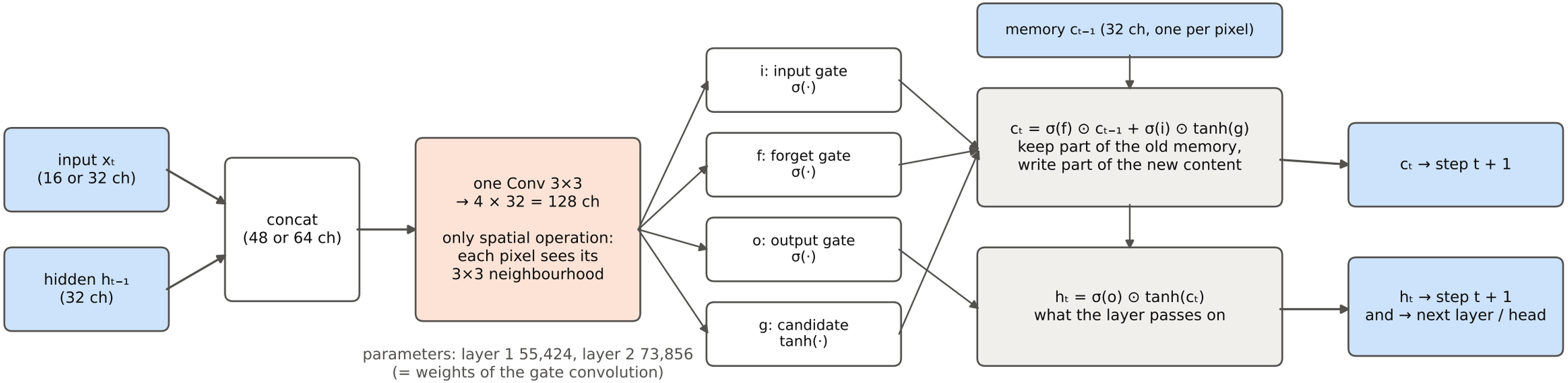
Inside the ConvLSTM and how the receptive field grows

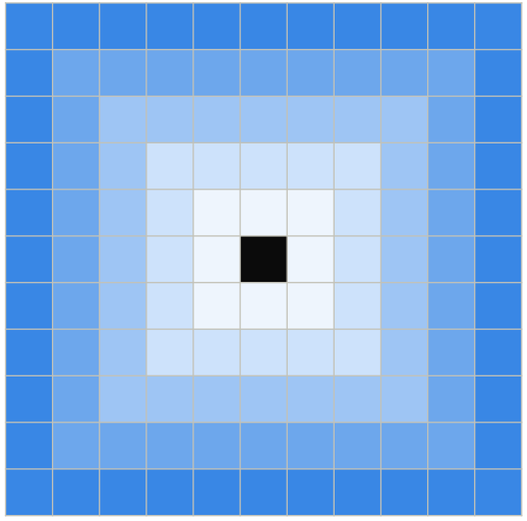
(a) One ConvLSTM cell at time step t (layer 1: x = 16 encoder channels; layer 2: x = 32 channels of layer 1)



parameters: layer 1 55,424, layer 2 73,856
(= weights of the gate convolution)

⊙ = element-wise per pixel: no exchange between pixels

(b) Within one time step: every 3×3 convolution adds one pixel in each direction



black = output pixel; one pixel ≈ 3.2 × 4.5 km

encoder conv 1: 3 × 3 px

encoder conv 2: 5 × 5 px

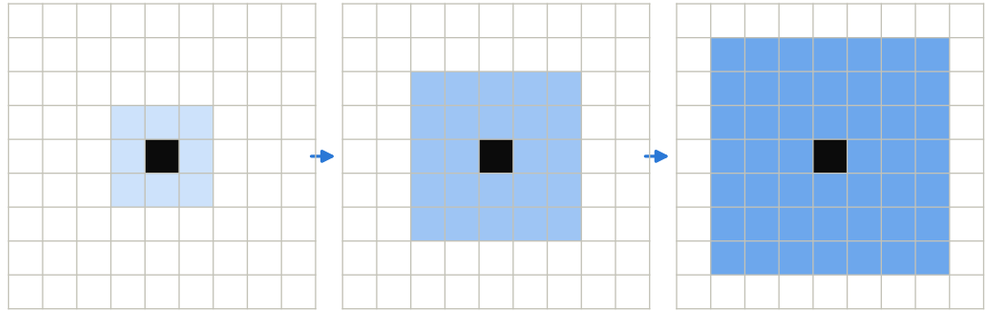
ConvLSTM 1 gates: 7 × 7 px

ConvLSTM 2 gates: 9 × 9 px

head conv 3×3: 11 × 11 px

9 × 9 px ≈ 29 × 40 km after layer 2
(only 7 × 7 with a single ConvLSTM layer)

(c) Over time: h_{t-1} passes the 3×3 convolution again each step



step t: 3 × 3 step t + 1: 5 × 5 step t + 2: 7 × 7

information from a neighbour arrives one step later, from a neighbour's neighbour two steps later …
in theory the whole crop is reachable after 24 steps; in practice the gates damp distant / old
signals, so the effective receptive field is much smaller (motivation for attention over time)